Module 4 Practice Problems: Worked Solutions

Mathematical Foundations for Data Science & Computational Intelligence · Mohsin Raza · Fall 2026

Try each problem before reading its solution. Every number here was checked by computing it. On an exam, method earns marks even when the arithmetic slips, so show the steps, not just the final answer.

Part A

Problem 1 Solvable or not?

(a) The first two equations, $x_1 + x_2 = 2$ and $x_1 + 2x_2 = 3$, give $x_2 = 1$ and $x_1 = 1$. The third equation needs $x_1 + 3x_2 = 4$: indeed $1 + 3 = 4$ ✓. So $\mathbf{x} = (1, 1)$ solves it exactly.

(b) The first two equations, $x_1 + x_2 = 1$ and $x_1 + 2x_2 = 3$, give $x_2 = 2$ and $x_1 = -1$. The third would need $x_1 + 3x_2 = 4$, but $-1 + 6 = 5 \neq 4$. **No solution.**

(c) $A\mathbf{x} = x_1(1, 1, 1) + x_2(1, 2, 3)$ is always a combination of A's two columns, which span a plane in $\mathbb{R}^3$. $\mathbf{b}_1$ lies in that plane (it's $1{\cdot}\text{col}_1 + 1{\cdot}\text{col}_2$); $\mathbf{b}_2$ does not. Rank test: $\text{rank}(A) = 2$ in both cases. $\text{rank}([A \,|\, \mathbf{b}_1]) = 2$ (adding $\mathbf{b}_1$ adds no new direction), so the system is solvable. $\text{rank}([A \,|\, \mathbf{b}_2]) = 3$ (the rank goes up), so it is not.

Problem 2 How many solutions?

	Exact solutions	$\hat{\mathbf{x}}$ unique?	Why
(a)	exactly one	yes	square, full rank: A is invertible
(b)	usually none	yes	tall: 100 equations, 4 unknowns; independent columns make $A^\top A$ invertible
(c)	infinitely many	no	wide: 3 equations, 5 unknowns, rank 3, so every $\mathbf{b}$ is reachable with 2 free directions left over
(d)	usually none	no	rank $2 < 3$ columns: many $\mathbf{x}$ give the same best fit; the pseudoinverse picks the shortest

In every case the *fitted values* $A\hat{\mathbf{x}}$ are unique: they're the projection of $\mathbf{b}$ onto the column space. What can fail to be unique is the *coefficient vector* $\hat{\mathbf{x}}$, exactly when the columns of A are dependent (cases (c) and (d)).

Part B

Problem 3 A least-squares line by hand

(a) $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\quad \mathbf{b} = (1, 2, 4, 5)$.

(b) $A^\top A = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$, $\quad A^\top \mathbf{b} = \begin{bmatrix} 1+2+4+5 \\ 0+2+8+15 \end{bmatrix} = \begin{bmatrix} 12 \\ 25 \end{bmatrix}$.

Solve $4x_1 + 6x_2 = 12$ and $6x_1 + 14x_2 = 25$. Multiply the first by $3/2$: $6x_1 + 9x_2 = 18$. Subtract: $5x_2 = 7$, so $x_2 = 1.4$. Then $4x_1 = 12 - 8.4 = 3.6$, so $x_1 = 0.9$.

$$\hat{\mathbf{x}} = (0.9,\ 1.4), \qquad t \approx 0.9 + 1.4\,s.$$

(c) Fitted values (0.9, 2.3, 3.7, 5.1); residuals $\mathbf{r} = (0.1, -0.3, 0.3, -0.1)$.
Column 1 (ones): $0.1 - 0.3 + 0.3 - 0.1 = 0$ ✓. Column 2: $0(0.1) + 1(-0.3) + 2(0.3) + 3(-0.1) = 0$ ✓.

(d) $0.9 + 1.4 \cdot 4 = 6.5$.

Problem 4 Projection onto a single vector

(a) With one column, $A^\top A\,\hat{x} = A^\top \mathbf{b}$ becomes $(\mathbf{a}^\top \mathbf{a})\,\hat{x} = \mathbf{a}^\top \mathbf{b}$:

$$\hat{x} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} = \frac{3 + 0 + 12}{1 + 4 + 4} = \frac{15}{9} = \frac{5}{3}.$$

(b) $\mathbf{p} = \frac{5}{3}(1, 2, 2) = \left(\frac{5}{3}, \frac{10}{3}, \frac{10}{3}\right)$, $\mathbf{r} = \mathbf{b} - \mathbf{p} = \left(\frac{4}{3}, -\frac{10}{3}, \frac{8}{3}\right)$.
Check: $\mathbf{r}^\top \mathbf{a} = \frac{4}{3} - \frac{20}{3} + \frac{16}{3} = 0$ ✓.

(c) The distance is $\|\mathbf{r}\| = \frac{1}{3}\sqrt{16 + 100 + 64} = \frac{\sqrt{180}}{3} = 2\sqrt{5} \approx 4.47$.

Problem 5 The projection matrix

(a) $\mathbf{a}^\top \mathbf{a} = 2$, so $P = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

(b) $P^2 = \frac{1}{4}\begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = P$ ✓, and P is visibly symmetric ✓. $P^2 = P$: projecting something that's already on the line leaves it unchanged (the shadow of a shadow is itself). $P^\top = P$: the projection is *orthogonal*, dropping straight down rather than at an angle.

(c) $P\mathbf{b} = \frac{1}{2}(4, 4, 0) = (2, 2, 0)$; $(I - P)\mathbf{b} = (3, 1, 4) - (2, 2, 0) = (1, -1, 4)$.
Perpendicular: $2(1) + 2(-1) + 0(4) = 0$ ✓. Sum: $(2, 2, 0) + (1, -1, 4) = (3, 1, 4) = \mathbf{b}$ ✓.

Problem 6 Reading residuals

(a) $A^\top \mathbf{r} = \mathbf{0}$ says $\mathbf{r}$ is perpendicular to every column of A. The first column is all ones, and $(1, 1, 1, 1)\cdot\mathbf{r}$ is just the sum of the residuals. So the sum must be zero. This holds for *every* least-squares fit that includes an intercept.

(b) No: $\frac{4}{3} - \frac{10}{3} + \frac{8}{3} = \frac{2}{3} \neq 0$. There's no column of ones in that model, so nothing forces the sum to zero. $\mathbf{r}$ only has to be perpendicular to $\mathbf{a} = (1, 2, 2)$, and it is.

(c) No. The new column space contains the old one, so the old best point is still available. The closest point in a bigger space can only be as close or closer: $\|\mathbf{r}\|^2$ stays the same or goes down. (That's training error. Test error can go up: overfitting.)

Part C

Problem 7 A tiny change, a big difference

(a) Subtract the first equation from the second: $0.001\,x_2 = 0.001$, so $x_2 = 1$ and $x_1 = 1$. $\mathbf{x} = (1, 1)$.

(b) Now $0.001\,x_2 = 0.002$, so $x_2 = 2$ and $x_1 = 0$. $\mathbf{x} = (0, 2)$. A change of 0.001 in the data moved the solution by 1 in each entry, about a thousand times bigger.

(c) The columns $(1, 1)$ and $(1, 1.001)$ are almost parallel. Telling apart "how much of each column" depends on a tiny difference between them, so tiny changes in $\mathbf{b}$ swing the answer wildly.

(d) Up to about $\log_{10}(4 \times 10^3) \approx 3.6$, so 3 to 4 digits. (It's a worst case: you often lose fewer, but you can't count on it.)

Problem 8 Condition numbers and the normal equations

(a) The singular values are $20,\ 0.5,\ 0.002$ (absolute values), so $\kappa = 20/0.002 = 10^4$.

(b) $\kappa(A^\top A) = \kappa(A)^2 = 10^8$. Normal equations: lose up to about 8 digits, so at least about 8 survive. QR works with A itself: lose up to about 4, so at least about 12 survive. These are rough worst-case estimates; in practice you often lose fewer digits, but the normal equations are always the riskier route.

(c) Over 58–72 inches, h, h^2 and h^3 all increase almost in lockstep, so the columns are nearly parallel and $\kappa(A)$ is huge. The fix is to **standardize** first: $z = (h - \bar{h})/s_h$. Then z runs from about -1.6 to 1.6, and z^2 (U-shaped) and z^3 (S-shaped) point in genuinely different directions. On the real data, $\kappa(A)$ at degree 3 drops from about 10^9 to about 6. *Also accepted:* switching to a better-conditioned polynomial basis (e.g. orthogonal polynomials). That wasn't covered in class, but it's a correct fix.

Part D

Problem 9 QR by hand

(a) $\|\mathbf{a}_1\| = 2$, so $\mathbf{q}_1 = \frac{1}{2}(1,1,1,1)$.

(b) $\mathbf{a}_2^\top \mathbf{q}_1 = \frac{1}{2}(0+1+2+3) = 3$, so the projection is $3\,\mathbf{q}_1 = (1.5, 1.5, 1.5, 1.5)$.

$$\mathbf{v} = \mathbf{a}_2 - 3\mathbf{q}_1 = (-1.5,\ -0.5,\ 0.5,\ 1.5).$$

This is s minus its mean (1.5): **centering**. Its length is $\|\mathbf{v}\| = \sqrt{2.25 + 0.25 + 0.25 + 2.25} = \sqrt{5}$, so $\mathbf{q}_2 = \mathbf{v}/\sqrt{5}$.

(c) $R = \begin{bmatrix} \|\mathbf{a}_1\| & \mathbf{a}_2^\top \mathbf{q}_1 \\ 0 & \|\mathbf{v}\| \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & \sqrt{5} \end{bmatrix}$.

(d) $\mathbf{q}_1^\top \mathbf{b} = \frac{1}{2}(1+2+4+5) = 6$; $\quad \mathbf{q}_2^\top \mathbf{b} = \frac{1}{\sqrt{5}}(-1.5 - 1 + 2 + 7.5) = \frac{7}{\sqrt{5}}$.
Back-substitute, starting from the last row: $\sqrt{5}\,x_2 = 7/\sqrt{5}$, so $x_2 = 7/5 = 1.4$. Then $2x_1 + 3(1.4) = 6$, so $x_1 = 0.9$.
$\hat{\mathbf{x}} = (0.9,\ 1.4)$: the same as Problem 3, without ever forming $A^\top A$.

Problem 10 When the columns are already orthonormal

(a) Orthonormal columns mean $Q^\top Q = I$. So $Q^\top Q\,\hat{\mathbf{x}} = Q^\top \mathbf{b}$ becomes $\hat{\mathbf{x}} = Q^\top \mathbf{b}$: just dot products, nothing to invert.

(b) $\hat{\mathbf{x}} = \left(\mathbf{q}_1^\top \mathbf{b},\ \mathbf{q}_2^\top \mathbf{b}\right) = \left(\frac{3+1}{\sqrt{2}},\ 4\right) = (2\sqrt{2},\ 4)$.
$Q\hat{\mathbf{x}} = 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1,0) + 4(0,0,1) = (2,2,4)$. (Residual $(1,-1,0)$, which is perpendicular to both columns ✓.)

Part E

Problem 11 Spot the redundancy

All four are rank-deficient. Writing $\mathbf{1}$ for the intercept column:

(a) USD $= \frac{1}{\text{rate}} \times$ PKR: one column is a multiple of the other.

(b) °F $= 1.8 \times$ °C $+ 32 \times \mathbf{1}$: not a multiple, but a combination *with the intercept*.

(c) red + green + blue $= \mathbf{1}$: every row has exactly one 1. (This is why one category is usually dropped as the "baseline".)

(d) age + birth year $= 2026 \times \mathbf{1}$: the same trick as (b).

Predictions are unaffected: the column space, and so the projection, is the same as without the redundant column. **Coefficients** are not meaningful: infinitely many coefficient vectors give the same fit, so the effect is split arbitrarily. R's `lm()` reports `NA` for one coefficient; NumPy and scikit-learn silently pick one split.

Problem 12 The pseudoinverse by hand

(a) $A^\top A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$ has determinant $25 - 25 = 0$, so it has no inverse. The columns of A are identical.

(b) This is Problem 4 again with $\mathbf{a} = (1, 2)$: $c = \dfrac{(1,2)\cdot(1,3)}{(1,2)\cdot(1,2)} = \dfrac{7}{5} = 1.4$. The fitted values are (1.4, 2.8).

(c) Any $\mathbf{x}$ with $x_1 + x_2 = 1.4$ works: for example $(0.7, 0.7)$, $(1.4, 0)$ or $(2, -0.6)$. All best-fit vectors lie on the line $x_1 + x_2 = 1.4$. The shortest one is the point on that line closest to the origin: the foot of the perpendicular from the origin. The line's normal direction is $(1, 1)$, so that point is $t(1, 1)$ with $2t = 1.4$, i.e. $\mathbf{x} = (0.7, 0.7)$, length $0.7\sqrt{2} \approx 0.99$. For comparison: $\|(1.4, 0)\| = 1.4$ and $\|(2, -0.6)\| \approx 2.09$, both longer.

(d) $A^\top A$ has eigenvalue $\lambda_1 = 10$ with $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$, and $\lambda_2 = 0$ with $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$. So $\sigma_1 = \sqrt{10}$ and $\sigma_2 = 0$; the zero is dropped rather than inverted.
$\mathbf{u}_1 = A\mathbf{v}_1/\sigma_1 = \dfrac{(2,4)/\sqrt{2}}{\sqrt{10}} = \dfrac{(1,2)}{\sqrt{5}}$.

$$A^+ = \frac{1}{\sigma_1}\,\mathbf{v}_1\mathbf{u}_1^\top = \frac{1}{\sqrt{10}}\cdot\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{5}}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix} = \frac{1}{10}\begin{bmatrix}1 & 2\\1 & 2\end{bmatrix}, \qquad A^+\mathbf{b} = \frac{1}{10}\begin{bmatrix}7\\7\end{bmatrix} = \begin{bmatrix}0.7\\0.7\end{bmatrix}.$$

This matches the shortest solution from (c): the pseudoinverse returns the minimum-length least-squares solution.

Summary of the three solvers. Normal equations: fine by hand and for small, well-conditioned problems; squares κ; fail on redundant columns. QR: the default (R's `lm()`); works with $\kappa(A)$; detects redundancy. SVD / pseudoinverse: handles redundant columns (minimum-length solution); most expensive.